

A Pharmacy Patient Case Presentation: Initial Intravenous Vancomycin Dosing for Sepsis in a Peritoneal Dialysis Patient



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Objectives

1. Review the diagnostic criteria and pathophysiology of sepsis.
2. Explain how to classify the severity of sepsis
3. Understand vancomycin dosing in peritoneal dialysis (PD)

Introducing the patient



Patient RA

- Age: 46
- Gender: Female
- Allergies: amoxicillin, benzonatate, nuts and strawberries.
- Weight: 100.4 kg
- BMI : 38
- Height: 162.5 cm
- **Past Medical History:**
 - End stage renal disease on peritoneal dialysis
 - Obesity



Pertinent Information

Chief Complaint: Cough, fever and lightheadedness

HPI: 46 y/o female presents to the emergency department with cough, fever, tachycardia and lightheadedness. Initial findings:

- BP 66/43 mmHG
- SpO₂ at 85% on room air
- Temp 102.4 °F
- Lactic Acid: 3.4
- Physical exam: Labial lesions

Cultures:

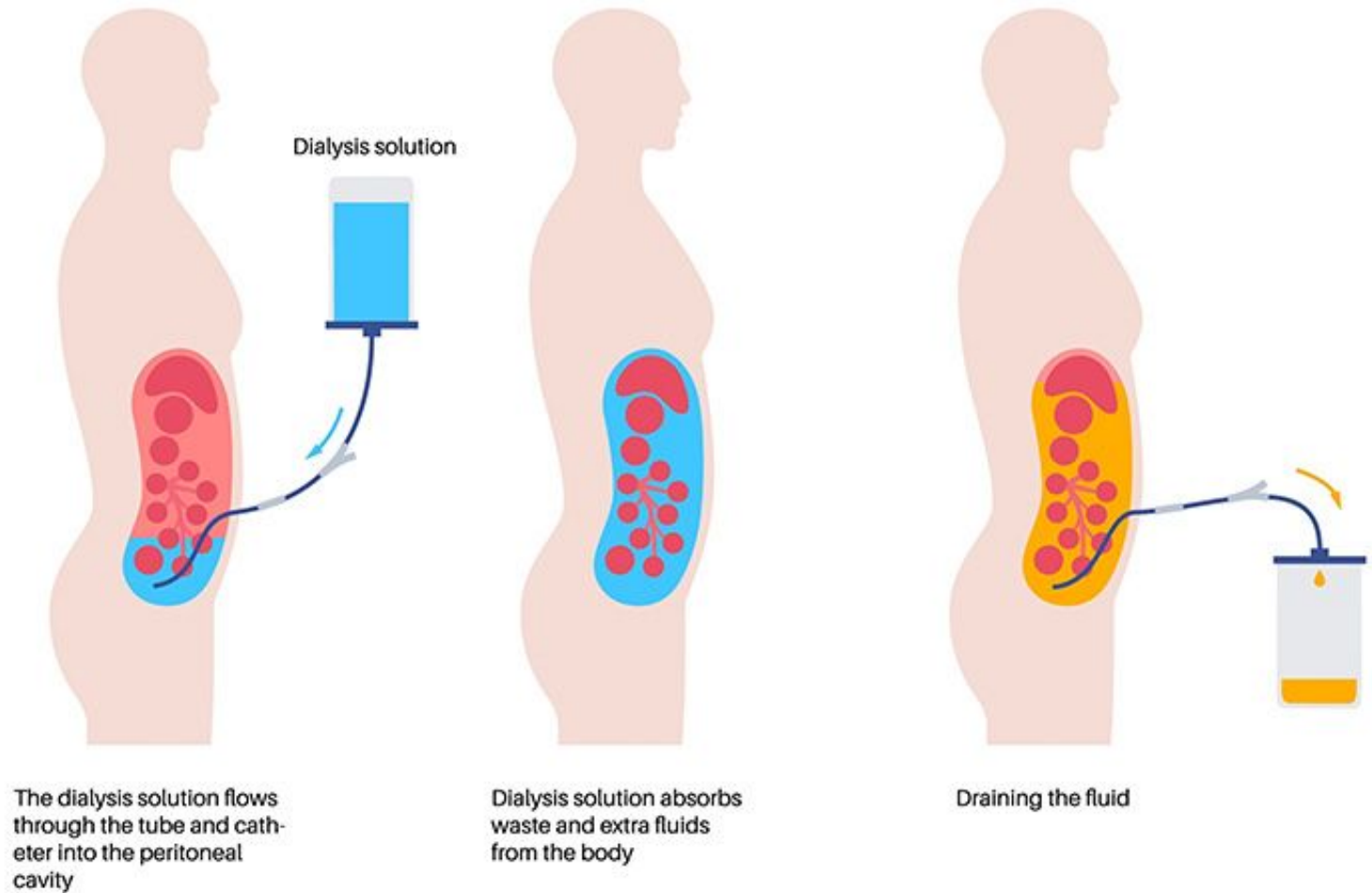
- Blood: no growth
- Peritoneal fluid: no growth
- Vaginal lesion: moderate gram positive cocci organisms, growth does not include S.aureus or beta hemolytic streptococcus.



Peritoneal Dialysis



Peritoneal Dialysis



Introducing Sepsis



Etiology

- Some of the most common sources of sepsis includes respiratory, gastrointestinal, genitourinary and skin or soft tissue infections, accounting for >80% of all sepsis cases.
- Pneumonia is the most common cause of sepsis.
- Gram negative (62%) or gram positive (47%) are the most common organisms identified in sepsis.
- Respiratory tract infections are more likely to be culture-negative, whereas urinary tract infection are likely to be culture-positive .

Sepsis

- Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection.
- Management of sepsis focuses on addressing the source of the infection and restoring hemodynamic stability

 <u>Sepsis</u> Suspected Infection + 2 SIRS below	 <u>Severe Sepsis</u> Sepsis + 1 Organ Dysfunction below	 <u>Septic Shock</u> Severe Sepsis + 1 below
☐ T < 96.8 or > 100.4 ☐ HR > 90 ☐ RR > 20 ☐ WBC > 12,000 or < 4,000 or bands > 10% *Consider sepsis for new or unexplained mental status change	☐ Lactate > 2 ☐ SBP < 90 or MAP < 65 ☐ SBP decrease > 40 from baseline ☐ Creatinine > 2 ☐ Urine output < 0.5 ml/kg/hr. for 2 hrs. ☐ Bilirubin > 2 ☐ Platelet count < 100 ☐ Coagulopathy (INR > 1.5 or aPTT > 60) ☐ Respiratory failure requiring ventilation	☐ Lactate ≥ 4 ☐ Sustained Hypotensive SBP < 90, despite fluid resuscitation 30mL/kg ☐ MAP < 65, despite fluid resuscitation 30mL/kg ☐ Vasopressor support required

Types of Shock

Hypovolemic Shock	Cardiogenic Shock	Obstructive Shock	Distributive/Septic Shock
• Loss of intravascular volume, mostly due to hemorrhage or severe dehydration.	• Caused by cardiac dysfunction such as acute myocardial infarction or severe heart failure, leading to a reduced cardiac output.	• Caused by physical obstruction of the heart or vessels such as in pulmonary embolism, cardiac tamponade, blocking blood flow.	• Abnormal distribution of blood flow due to vasodilation leading to decreased blood pressure and poor perfusion of organs and tissues.

Diagnosing Sepsis

Guidelines emphasize on the importance of early identification of patients who may develop sepsis as a way to decrease sepsis related mortality

- SIRS (Systemic Inflammatory Response Syndrome) criteria
- qSOFA (quick Sepsis Related Organ Failure Assessment) score
- NEWS (National Early Warning Signs) score

Systemic Inflammatory Response Syndrome (SIRS) Criteria

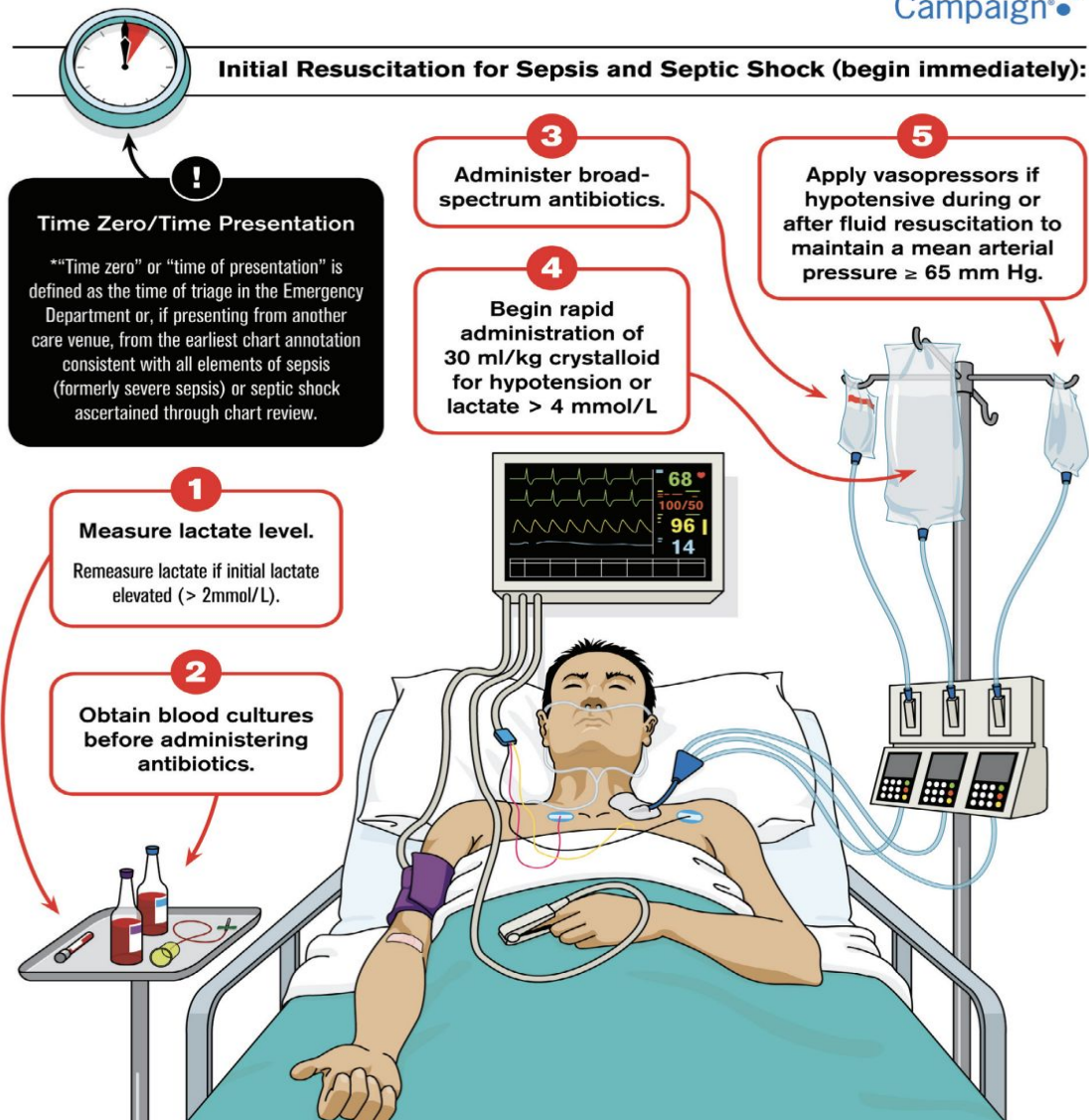
- SIRS is a clinical syndrome that is a state of dysregulated systemic inflammation.
- SIRS criteria was previously used by clinicians to identify sepsis, but is now less favored because it poorly predicts death from sepsis and is frequently met by hospitalized patients who do not have an infection.
- SIRS is defined as the presence of at least two of the following:
 - Temperature $>100.4^{\circ}\text{F}$ or $<96.8^{\circ}\text{F}$
 - Heart rate > 90 bpm
 - Respiratory rate >20 bpm or pp of carbon dioxide > 32 mmHg
 - White blood cell count $> 12,000$ per μL or < 4000 per μL or greater than 10% immature forms

qSOFA VS NEWS

qSOFA (quick Sequential Organ Failure Assessment) score	NEWS (New Early Warning) score
Modified version of Sequential Organ Failure Assessment score (SOFA). A score of more than 2 is linked to poor outcomes due to sepsis. qSOFA only has three components: • Respiratory rate ≥ 22/minute • Altered mental status • Systolic blood pressure ≤ 100 mmHg	Aggregate scoring derived from six physiologic parameters: • Respiratory rate • Oxygen saturation • Systolic blood pressure • Temperature • Pulse rate • Level of consciousness or new confusion. The score represents risk of death from sepsis and indicates urgency. 0-4 - low risk 5-6 - medium risk 7 or more - high risk

Treatment Course of Sepsis

Surviving Sepsis
Campaign



Back To Patient RA

Patient RA - Presentation

- BP 66/43 mmHG
- SpO₂ at 85% on room air
- Temp 100.2 °F
- Heart Rate 108 bpm
- Lactic Acid: 3.4
- Physical exam: Labial lesions

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ED Medications

- Cefepime 2 gm x1 LD
- Vancomycin 2000 mg LD
- Sodium chloride 0.9 % IV bolus (30 mL/kg) per protocol
- Norepinephrine drip
- 2 L of Oxygen

Patient RA - After Initial Treatment

- SPO2 97% on room air
- Normal CBC (including platelet)
- No evidence of liver dysfunction
- BP 120/63 mHG
- Creatinine of 16.7 mg/dL
- No AMS
- Lactic: 1.3

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Vancomycin in PD

Intraperitoneal administration is typically reserved for peritoneal infection in dialysis patients and is NOT intended for systemic infections.

- The mean elimination half life of PD can be up to 7.5 days vs 4-6 hours in those with normal renal function.
- PD removes about 28% of vancomycin
- Dosing intervals are typically every 48-72 hours
- Hemodialysis is usually more efficient than PD at clearing vancomycin

Vancomycin in PD

- Vancomycin dosing recommendation for PD
 - LD of 20 mg/kg TBW
 - Vancomycin goal of 15-20 mcg/mL
 - Random level within 72 hours after initial dosing
 - If level $\leq$ 20 mcg/mL - administer 15 mg/kg every 4 -7 days
 - If level $>$ 20 mcg/mL - hold dose and repeat level every 24 hours until less than 20 mcg/mL
 - If level within range - repeat level every 24-48 hours

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Patient RA's Vancomycin Course

- Continuous PD at home with last session completed prior to admission without issues
- 2/8: 2000 mg x 1 LD
- 2/9: PD held due to patient discomfort
- 2/10: Random Level: 25.8 at 0808
- 2/10: 2 hours of HD at 1636
- 2/10: 1000 mg x1 scheduled
- 2/11: Random Level: 31.4 at 0916
- 2/11: Vancomycin dose held
- 2/11: Vancomycin discontinued
- 2/12: 3 hour of HD at 1300
- 2/12: Doxycycline started

Vancomycin Regimen Assessment

- The first random vancomycin level drawn **48 hours** after LD was supratherapeutic at 25.8 mcg/mL
 - **Additional 1g vancomycin dose was given ordered**
- 24 hours after a second random level was 31.4 mcg/mL (supratherapeutic)
- Vancomycin dose was held.

Conclusions and Key Takeaways

- The recommended initial vancomycin loading dose for systemic infection is 20 mg/kg (total body weight).
- Maintenance dose should be individualized and guided by trough concentrations.
- Dosing interval is typically every 4 to 7 days.
- Adjustment should be based on trough levels and clinical status, requiring early and frequent monitoring.

Questions?



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